

Week3 Practical

Task 1) Document Modeling — create a `book_metadata` collection with embedded patterns (reviews list inside the book document) and referenced patterns (linking to `member_id` in SQL). 2) Advanced Querying — use comparison operators (`$gt`, `$in`) to find books rated above 4, and a regex query to find books with "Digital" formats. 3) Aggregation Pipeline — match books published after 2020, `$unwind` the reviews array, then `$group` by title and compute the average rating. 4) Indexing for Speed — create a text index on the review comments field and use `.explain()` to check whether the query uses the index.

1) Document Modeling

The screenshot shows the MongoDB BookFlow application interface. The breadcrumb navigation at the top indicates the path: Student > BookFlow > book_metadata. A button labeled "Open MongoDB shell" is in the top right corner. Below the breadcrumb, there are tabs for "Documents" (with a count of 4), "Aggregations", "Schema", "Indexes" (with a count of 1), and "Validation". The "Documents" tab is active. A search bar contains the placeholder text "Type a query: { field: 'value' } or Generate query" and buttons for "Explain", "Reset", "Find", and "Options". Below the search bar, there are buttons for "ADD DATA", "BULK", "EXPORT DATA", and "EXPORT CODE". A pagination bar shows "25" items per page and "1 - 4 of 4" documents. The main area displays two document snippets. The first document is for "The Silent Algorithm" by R. Menon, published in 2022, with an average rating of 4.6. The second document is for "Ocean of Code" by A. Fernandes, published in 2021, with an average rating of 4.2. Both documents have embedded arrays for "formats", "specs", and "reviews".

```
_id: ObjectId('6a8d08239998e4beab722eca')
title: "The Silent Algorithm"
author: "R. Menon"
genre: "Technology"
publishedYear: 2022
formats: Array (3)
specs: Object (5)
avgRating: 4.6
reviews: Array (2)
```

```
_id: ObjectId('6a8d08309998e4beab722ecc')
title: "Ocean of Code"
author: "A. Fernandes"
genre: "Technology"
publishedYear: 2021
formats: Array (2)
specs: Object (3)
avgRating: 4.2
reviews: Array (2)
```

```
book_metadata | mongosh: Student +
>_MONGOSH
> use("BookFlow")
< switched to db BookFlow
> db["book_metadata"].find()
< {
  _id: ObjectId('6a8d08239998e4beab722eca'),
  title: 'The Silent Algorithm',
  author: 'R. Menon',
  genre: 'Technology',
  publishedYear: 2022,
  formats: [
    'Hardcover',
    'Digital',
    'Audiobook'
  ],
  specs: {
    pages: 340,
    fileSize_MB: 4.2,
    drm: true,
    narrator: 'K. Rao',
    runtime_min: 610
  },
  avgRating: 4.6,
  reviews: [
    {
      member_id: 'M1001',
```

Task2: Advanced Querying

```
{
  "avgRating": {
    "$gt": 4
  }
}
```

[Generate query](#) [Explain](#) [Reset](#) [Find](#) [Options](#)

[+ ADD DATA](#) [BULK](#) [EXPORT DATA](#) [EXPORT CODE](#) [?](#)

25 1 - 3 of 3 [↺](#) [↻](#) [↷](#) [☰](#) [🔍](#) [📄](#)

```
_id: ObjectId('6a8d08239998e4beab722eca')
title: "The Silent Algorithm"
author: "R. Menon"
genre: "Technology"
publishedYear: 2022
▶ formats: Array (3)
▶ specs: Object (5)
  avgRating: 4.6
▶ reviews: Array (2)
```

```
_id: ObjectId('6a8d08309998e4beab722ecc')
title: "Ocean of Code"
author: "A. Fernandes"
genre: "Technology"
publishedYear: 2021
▶ formats: Array (2)
▶ specs: Object (3)
  avgRating: 4.2
▶ reviews: Array (2)
```

Documents4

Aggregations

Schema

Indexes1

Validation

```
{
  "formats": {
    "$in": [
      "Digital",
      "Audiobook"
    ]
  }
}
```

Generate query

Explain

Reset

Find

Options

ADD DATA

BULK

EXPORT DATA

EXPORT CODE

251 - 3 of 3

```
_id: ObjectId('6a8d08239998e4beab722eca')
title: "The Silent Algorithm"
author: "R. Menon"
genre: "Technology"
publishedYear: 2022
formats: Array (3)
specs: Object (5)
avgRating: 4.6
reviews: Array (2)
```

```
_id: ObjectId('6a8d08309998e4beab722ecc')
title: "Ocean of Code"
author: "A. Fernandes"
genre: "Technology"
publishedYear: 2021
formats: Array (2)
```

Documents4

Aggregations

Schema

Indexes1

Validation

```
{
  "avgRating": {
    "$gt": 4
  },
  "formats": {
    "$in": [
      "Digital"
    ]
  }
}
```

Generate query

Explain

Reset

Find

Options

ADD DATA

BULK

EXPORT DATA

EXPORT CODE

251 - 3 of 3

```
_id: ObjectId('6a8d08239998e4beab722eca')
title: "The Silent Algorithm"
author: "R. Menon"
genre: "Technology"
publishedYear: 2022
formats: Array (3)
specs: Object (5)
avgRating: 4.6
reviews: Array (2)
```

```
_id: ObjectId('6a8d08309998e4beab722ecc')
title: "Ocean of Code"
author: "A. Fernandes"
genre: "Technology"
publishedYear: 2021
```


Task3: Aggregation Pipeline

\$match

\$unwind

\$group

\$sort

\$project

Generate aggregation Explain

Untitled - modified SAVE CREATE NEW EXPORT DATA EXPORT CODE PREVIEW STAGES

+

Stage 5

\$project

```
1 {
2   "_id": 0,
3   "title": "$_id",
4   "author": 1,
5   "avgReviewRating": {
6     "$round": [
7       "$avgReviewRating",
8       2
9     ]
10  },
11  "totalReviews": 1
12 }
```

Output preview after [\\$project](#) stage (Sample of 3 documents)

```
author : "R. Menon"
totalReviews : 2
title : "Quantum Horizons"
avgReviewRating : 5
```

\$match

\$unwind

\$group

\$sort

\$project

Edit Explain Run

ALL RESULTS EXPORT DATA EXPORT CODE Showing 1 - 3 count results

```
author : "R. Menon"
totalReviews : 2
title : "Quantum Horizons"
avgReviewRating : 5
```

```
author : "R. Menon"
totalReviews : 2
title : "The Silent Algorithm"
avgReviewRating : 4.5
```

```
author : "A. Fernandes"
totalReviews : 2
title : "Ocean of Code"
avgReviewRating : 4.5
```

Task4: Indexing for Speed

Student > BookFlow > book_metadata

Documents 4 Aggregations Schema **Indexes 2** Validation

Create Index Refresh

VIEWING INDEXES SEARCH INDEXES

Name & Definition	Type	Size	Usage	Properties	Status
> _id_	REGULAR ⓘ	20.5 kB	7 (since Tue Aug 25 2026)	UNIQUE ⓘ	READY
> reviews.comment_text	TEXT ⓘ	20.5 kB	0 (since Tue Aug 25 2026)		READY

Student > BookFlow > book_metadata

Documents 4 Aggregations Schema Indexes 2 Validation

⌵

{
 "\$text": {
 "\$search": "Digital"
 }
}

Generate query ⚡ Explain Reset Find Option

ADD DATA BULK EXPORT DATA EXPORT CODE

25 1 - 3 of 3

```
_id: ObjectId('6a8d08239998e4beab722eca')
title: "The Silent Algorithm"
author: "R. Menon"
genre: "Technology"
publishedYear: 2022
formats: Array (3)
specs: Object (5)
avgRating: 4.6
reviews: Array (2)
```

```
_id: ObjectId('6a8d08519998e4beab722ed0')
title: "Quantum Horizons"
author: "R. Menon"
genre: "Science"
publishedYear: 2023
formats: Array (2)
specs: Object (4)
avgRating: 4.9
reviews: Array (2)
```

Documents 4 Aggregations Schema **Indexes 3** Validation

Create Index Refresh

VIEWING INDEXES SEARCH INDEXES

Name & Definition	Type	Size	Usage	Properties	Status
> _id_	REGULAR ⓘ	20.5 kB	7 (since Tue Aug 25 2026)	UNIQUE ⓘ	READY
> reviews.comment_text	TEXT ⓘ	20.5 kB	6 (since Tue Aug 25 2026)		READY
> formats_1	REGULAR ⓘ	20.5 kB	0 (since Tue Aug 25 2026)		READY

Documents 4AggregationsSchemaIndexes 4Validation

Create IndexRefreshVIEWINGINDEXESSEARCH

Name & Definition	Type	Size	Usage	Properties	Status
> _id_	REGULAR	20.5 kB	7 (since Tue Aug 25 2026)	UNIQUE	READY
> reviews.comment_text	TEXT	20.5 kB	6 (since Tue Aug 25 2026)		READY
> formats_1	REGULAR	20.5 kB	0 (since Tue Aug 25 2026)		READY
> avgRating_1	REGULAR	20.5 kB	0 (since Tue Aug 25 2026)		READY

Documents 4AggregationsSchemaIndexes 4Validation

{
 "avgRating": {
 "\$gt": 4
 }
}

Generate query +

Explain

Reset

Find

ADD DATA

BULK

EXPORT DATA

EXPORT CODE

25

1 - 3 of 3

```
_id: ObjectId('6a8d08519998e4beab722ed0')  
title: "Quantum Horizons"  
author: "R. Menon"  
genre: "Science"  
publishedYear: 2023  
formats: Array (2)  
specs: Object (4)  
avgRating: 4.9  
reviews: Array (2)
```

```
_id: ObjectId('6a8d08239998e4beab722eca')  
title: "The Silent Algorithm"  
author: "R. Menon"  
genre: "Technology"  
publishedYear: 2022  
formats: Array (3)  
specs: Object (5)  
avgRating: 4.6  
reviews: Array (2)
```